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Course Curriculum

Generative AI With Agentic AI Masters Program Certification

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- Benjamin Franklin

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- Continuous Integration, Deployment & Scaling LLMs((CI/CD & Scaling)

1

Python & Statistics Essentials for Gen AI and Machine Learning

Session 1: Python Programming Fundamentals

Goal: Build strong foundational skills in Python syntax, operations, and interactive coding.

Topics:

- Introduction to Python
 - Why Python: simplicity, readability, and ecosystem.
 - Installing Python & IDE setup (Jupyter/VS Code).
 - Running Python scripts vs interactive shell.
- Basic Syntax & Data Types
 - Variables, constants, and naming conventions.
 - Primitive types: integers, floats, strings, booleans.
 - Type casting and type checking.
- Operators & Expressions
 - Arithmetic, comparison, logical, and assignment operators.
 - String operations and f-strings.

Hands-On Activity

- Write a script that takes user input and performs arithmetic operations.
- Implement a small calculator using Python operators.

Session 2: Data Structures, Control Flow, and File Operations

Goal: Learn to organize data, control program flow, and work with files.

Topics:

- Data Structures in Python
 - Lists, tuples, sets, and dictionaries.
 - Indexing, slicing, and mutability concepts.
 - Common operations: append, extend, pop, update.
- Control Flow
 - Conditional statements (if, elif, else).
 - Loops: for, while, and break/continue/pass.
 - List comprehensions for concise loops.
- File Handling
 - Opening, reading, writing, and closing files.
 - File modes: text vs binary.
 - Using with for safe file operations.

Hands-On Activity

- Parse a log file and count word frequencies.
- Write results into a new file.

1

Python & Statistics Essentials for Gen AI and Machine Learning

Session 3: Functions, Object-Oriented Programming, and Exception Handling

Goal: Master reusable code with functions, OOP principles, and error handling.

Topics:

- Functions
 - Defining and calling functions.
 - Arguments: positional, keyword, default, and *args/**kwargs.
 - Scope and return values.
- Object-Oriented Programming (OOP)
 - Sessions and objects in Python.
 - Attributes, methods, and `__init__`.
 - Inheritance, polymorphism, and encapsulation.
- Exception Handling
 - Common Python errors and exceptions.
 - Using try, except, finally, and raise.
 - Custom exceptions.

Hands-On Activity

- Create a BankAccount Session with deposit/withdraw methods.
- Handle errors such as insufficient balance using exceptions.

Session 4: Data Analysis and Visualization

Goal: Analyze datasets and visualize insights using Python libraries.

Topics:

- Data Analysis with Pandas
 - Series and DataFrames.
 - Importing CSV/Excel data.
 - Filtering, grouping, and aggregating.
- Numerical Computing with NumPy
 - Arrays and vectorized operations.
 - Broadcasting and reshaping.
 - Mathematical/statistical functions.
- Data Visualization with Matplotlib & Seaborn
 - Line, bar, histogram, and scatter plots.
 - Styling and customization.
 - Seaborn for statistical visualization (heatmaps, pairplots).

Hands-On Activity

- Load a dataset (e.g., Titanic or Iris).
- Perform exploratory data analysis (EDA).
- Create 2–3 meaningful plots to summarize findings.

2

Applied Machine Learning

Session 1: Data Pre-processing and Feature Engineering

Goal: Prepare raw data for modeling by handling missing values, encoding variables, and creating meaningful features.

Topics:

- Data Cleaning & Pre-processing
 - Handling missing data: deletion, imputation (mean, median, mode, advanced methods).
 - Outlier detection and treatment (IQR, Z-score).
 - Data transformations: scaling (MinMax, Standard), normalization, log transforms.
- Feature Engineering
 - Encoding categorical variables: one-hot, label encoding, target encoding.
 - Feature creation: domain-driven features, interaction terms, polynomial features.
 - Feature selection: filter (correlation), wrapper (RFE), embedded (Lasso, tree-based).

Hands-On Activity

- Pre-process a dataset (e.g., Titanic).
- Create new engineered features and evaluate their impact.

Session 2: Supervised Learning – Regression

Goal: Learn regression techniques for prediction and forecasting.

Topics:

- Linear Regression
 - Assumptions (linearity, independence, homoscedasticity, normality)
 - OLS estimation, residual analysis, R^2 vs Adjusted R^2
 - Limitations in real-world datasets
- Regularized Regression
 - Ridge, Lasso, ElasticNet
 - Bias-variance tradeoff explained practically
 - Feature selection effect (Lasso shrinking coefficients)
- Polynomial & Interaction Terms
 - When linear models underfit
 - Tradeoff with overfitting
- Tree-based Regression Models
 - Decision Tree regression basics
 - Overfitting control (max_depth, min_samples_leaf)
 - Use case: predicting response time of a server

Hands-On Activity

- Predict server response time from CPU utilization & memory usage (Ops dataset).
- Compare Linear Regression vs Ridge vs Lasso vs Decision Tree.
- Visualize feature importance.

2

Applied Machine Learning

Session 3: Supervised Learning – Sessionification

Goal: Train Sessionification models with real-world applications.

Topics:

- Logistic Regression
 - Sigmoid function, decision boundary, interpretation of coefficients
 - When it fails (non-linear separation)
- K-Nearest Neighbors
 - Distance metrics (Euclidean, Manhattan, cosine)
 - Effect of K on bias/variance
- Decision Trees for Sessionification
 - Gini impurity vs entropy
 - Tree depth & pruning
 - Interpretability advantage
- Random Forests & Gradient Boosted Trees (XGBoost, LightGBM)
 - Ensemble basics (bagging vs boosting)
 - Why boosting dominates in Kaggle competitions
- Evaluation Metrics
 - Accuracy, Precision, Recall, F1, ROC-AUC
 - Confusion Matrix deep dive

Hands-On Activity

- Build a server anomaly Sessionifier (normal vs abnormal logs).
- Compare Logistic Regression, Random Forest, and XGBoost.
- Evaluate results using Precision/Recall/ROC-AUC.

Session 4: Unsupervised Learning

Goal: Discover hidden patterns in unlabeled data.

Topics:

- K-Means Clustering
 - Algorithm steps (centroid update, assignment)
 - Choosing K: Elbow method, Silhouette score
- Hierarchical Clustering
 - Agglomerative vs divisive clustering
 - Dendrogram interpretation
- Density-based Clustering
 - DBSCAN for arbitrary shapes & noise handling
 - Application in anomaly detection
- Dimensionality Reduction Intro
 - Why high dimensions hurt (curse of dimensionality)

2

Applied Machine Learning

- PCA basics (variance explained)

Hands-On Activity

- Cluster server usage patterns (group servers by utilization trends).
- Compare K-Means vs DBSCAN.
- Visualize clusters in 2D (after PCA reduction).

Session 5: Clustering Algorithms & Dimensionality Reduction

Goal: Go deeper into clustering + DR for high-dimensional ML.

Topics:

- Advanced Dimensionality Reduction
 - PCA (math + intuition)
 - t-SNE for visualization (non-linear)
 - UMAP (faster t-SNE alternative)
- Clustering in High Dimensions
 - Gaussian Mixture Models (soft clustering)
 - Applications: customer segmentation, anomaly detection
- Feature Importance with DR
 - Interpreting reduced dimensions
 - Tradeoff: interpretability vs performance

Hands-On Activity

- Apply PCA to high-dimensional e-commerce features (products, categories, transactions).
- Visualize with t-SNE/UMAP.
- Perform GMM clustering and interpret results.

Session 6: Ensemble Learning

Goal: Understand ensemble methods and their power in real ML systems.

Topics:

- Ensemble Concepts
 - Bagging, boosting, stacking
 - Why ensembles reduce variance/bias
- Bagging Algorithms
 - Random Forest deep dive
 - OOB (out-of-bag) error estimation
- Boosting Algorithms
 - AdaBoost (intuition)
 - Gradient Boosting
 - XGBoost, LightGBM, CatBoost

2

Applied Machine Learning

- Stacking
 - Combining multiple base learners
 - Meta-learning strategies
- Real-world Usage
 - Kaggle solutions (why ensembles dominate)
 - Industry examples (fraud detection, predictive maintenance)

Hands-On Activity

- Use e-commerce order dataset for predicting customer churn.
- Train Random Forest, XGBoost, CatBoost.
- Build a stacking ensemble → compare performance.



3

Deep Learning & Neural Network Architectures

Session 1: Neural Networks & Deep Learning Foundations

Goal: Build a strong foundation in neural network theory and mechanics of training deep learning models.

Topics:

- Biological inspiration vs Artificial Neural Networks.
- Mathematical formulation of a perceptron.
- Forward propagation mechanics: weighted sums & activations.
- Activation functions: sigmoid, tanh, ReLU, GELU, Softmax.
- Loss functions for Classification & regression (MSE, Cross-Entropy).
- Backpropagation & gradient descent explained.
- Optimizers: SGD, Adam, RMSProp.
- Vanishing & exploding gradients problem.

Hands-On Activity:

- Implement a single-layer perceptron in Python from scratch.
- Train a feedforward NN using PyTorch/TensorFlow on MNIST dataset.
- Visualize decision boundaries of simple Classification tasks.

Session 2: Tuning & Optimizing Deep Neural Networks

Goal: Learn strategies to optimize deep networks for convergence, stability, and generalization.

Topics:

- Hyperparameter tuning: learning rate, batch size, epochs.
- Weight initialization strategies (Xavier, He, Kaiming).
- Regularization techniques: L1/L2, Dropout, Early Stopping.
- Batch Normalization & Layer Normalization.
- Data augmentation to prevent overfitting.
- Advanced optimizers: AdamW, Nadam, Lookahead.
- Learning rate schedules & warm restarts (OneCycle, Cosine decay).
- Debugging underperforming models (loss not converging, overfitting).

Hands-On Activity:

- Tune hyperparameters using Optuna on a neural network.
- Compare different optimizers on CIFAR-10 dataset.
- Apply data augmentation and measure impact on test accuracy.

Session 3: Convolutional Neural Networks – I

Goal: Understand convolutional networks for image feature extraction and Classification.

Topics:

- Convolution operation fundamentals (kernels, filters, stride, padding).

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Deep Learning & Neural Network Architectures

- Feature extraction through low-level and high-level filters.
- Pooling layers (max, average, global).
- CNN architecture components (Conv → Pool → Dense).
- Receptive field concept and hierarchical representation.
- Weight sharing & translation invariance in CNNs.
- Popular CNN models overview (LeNet, AlexNet).
- Comparison of CNN vs fully connected networks on image data.

Hands-On Activity:

- Build a CNN from scratch for MNIST digit Sessionification.
- Visualize feature maps and convolution filters.
- Compare CNN vs fully connected performance

Session 4: Convolutional Neural Networks – II

Goal: Dive into advanced CNN architectures and techniques for large-scale image Sessionification.

Topics:

- Deeper CNNs: VGG, ResNet, DenseNet.
- Skip connections & residual learning.
- Bottleneck layers & efficiency improvements.
- Transfer learning & fine-tuning pre-trained CNNs.
- Advanced layers: Depthwise Separable Convolutions (MobileNet).
- Regularization in CNNs: DropBlock, CutMix, Mixup.
- Object detection (YOLO, Faster R-CNN) introduction.
- CNNs in real-world applications (medical imaging, autonomous driving).

Hands-On Activity:

- Fine-tune ResNet on CIFAR-100 dataset.
- Visualize Grad-CAM for explainability in CNN predictions.
- Train and evaluate MobileNet for edge deployment.

3

Deep Learning & Neural Network Architectures

Self-paced: Long Short-Term Memory (LSTM) Networks

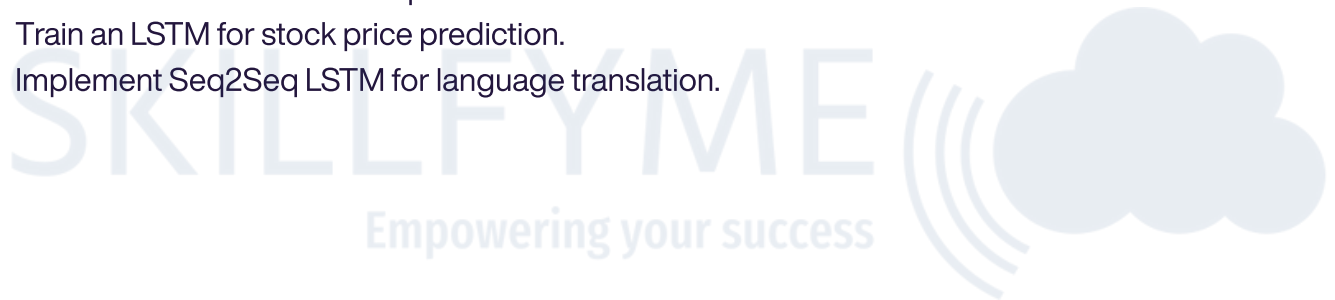
Goal: Explore advanced recurrent architectures for long-term sequence dependencies.

Topics:

- Motivation for LSTM: solving vanishing gradient problem.
- LSTM architecture: input, forget, and output gates.
- Cell state & hidden state dynamics.
- GRU vs LSTM trade-offs.
- Stacked LSTMs & sequence-to-sequence models.
- Attention mechanism introduction with LSTM.
- LSTMs for NLP tasks: text generation, machine translation.
- LSTMs for time series: anomaly detection, forecasting.

Hands-On Activity:

- Build an LSTM for next-word prediction.
- Train an LSTM for stock price prediction.
- Implement Seq2Seq LSTM for language translation.



4 Natural Language Processing (NLP)

Session 1: Introduction to NLP

Goal: Build foundational understanding of NLP, its applications, and challenges in working with human language.

Topics:

- Evolution of NLP: from rule-based → statistical → deep learning.
- Key applications: chatbots, machine translation, summarization.
- Text as unstructured data: challenges in processing.
- Bag-of-Words and TF-IDF basics.
- Language ambiguity: polysemy, synonyms, context.
- Evaluation metrics in NLP (BLEU, ROUGE, perplexity).
- Role of corpora & datasets (Wikipedia, IMDB, CoNLL).
- Limitations of Sessionical NLP vs neural approaches.

Hands-On Activity:

- Load and preprocess a text dataset using NLTK/spaCy.
- Build a simple Bag-of-Words model for document Sessionification.
- Compare word frequency distributions in different corpora.

Session 2: Text Processing & Feature Engineering

Goal: Master preprocessing and feature extraction for preparing raw text data for ML/DL models.

Topics:

- Text cleaning: lowercasing, punctuation removal, stopwords.
- Stemming vs Lemmatization.
- N-grams and collocations.
- Term weighting techniques (TF, IDF, TF-IDF).
- Word hashing & hashing trick.
- Sparse vs dense text representations.
- Handling out-of-vocabulary (OOV) words.
- Feature scaling & dimensionality reduction for text.

Hands-On Activity:

- Preprocess text dataset (IMDB reviews) using NLTK & spaCy.
- Extract TF-IDF features with scikit-learn.
- Train a logistic regression Sessionifier on processed text.

Session 3: Named Entity Recognition (NER) & Parsing

Goal: Extract entities and understand syntactic/semantic structures of text.

Topics:

- Introduction to Named Entity Recognition (NER).
- Common entity categories: person, organization, location.

4 Natural Language Processing (NLP)

- Rule-based vs statistical vs neural NER.
- Part-of-Speech (POS) tagging.
- Dependency parsing vs constituency parsing.
- Applications of parsing: relation extraction, question answering.
- Challenges in multilingual & domain-specific NER.
- Pre-trained NER models (spaCy, Hugging Face).

Hands-On Activity:

- Apply POS tagging & dependency parsing using spaCy.
- Train a custom NER model with spaCy on domain data.
- Visualize dependency trees and extracted entities.

Session 4: Tokenization & Text Encoding

Goal: Learn tokenization strategies and modern text embedding methods.

Topics:

- Word-level tokenization (NLTK, spaCy).
- Subword tokenization (BPE, WordPiece, SentencePiece).
- Character-level tokenization.
- One-hot encoding limitations.
- Distributed word embeddings: Word2Vec, GloVe, FastText.
- Contextual embeddings: ELMo, BERT.
- Embedding visualization (TSNE, UMAP).
- Encoding challenges: multilingual text, rare words.

Hands-On Activity:

- Compare word-level vs subword tokenization.
- Train Word2Vec embeddings on a custom text corpus.
- Use BERT embeddings for semantic similarity tasks.

Session 5: Sentiment Analysis Essentials

Goal: Build foundational sentiment classification models using ML.

Topics:

- Sentiment analysis overview and applications.
- Lexicon-based approaches.
- Traditional ML Classifiers: Naive Bayes, SVM, Logistic Regression.
- Feature engineering for sentiment tasks.
- Dataset considerations (imbalanced classes, sarcasm).
- Evaluation metrics (precision, recall, F1).
- Domain adaptation in sentiment analysis.
- Limitations of rule-based/ML-only sentiment models.

4 Natural Language Processing (NLP)

Hands-On Activity:

- Train a Naive Bayes Classifier on IMDB reviews.
- Compare SVM vs Logistic Regression for sentiment.
- Analyze misclassifications to identify improvements.

Session 6: Advanced Sentiment Analysis

Goal: Leverage deep learning and transformers for robust sentiment models.

Topics:

- RNNs (LSTM, GRU) for sentiment Classification.
- Attention mechanisms for contextual sentiment.
- Transfer learning with BERT for sentiment tasks.
- Fine-tuning transformers for domain-specific sentiment.
- Multi-Sentence sentiment Classification.
- Aspect-based sentiment analysis.
- Cross-lingual sentiment models.
- Explainability in sentiment Classification (LIME, SHAP).

Hands-On Activity:

- Train an LSTM for sentiment Classification.
- Fine-tune BERT on IMDB dataset.
- Perform aspect-based sentiment analysis with Hugging Face models

Session 7: Neural Language Models

Goal: Understand modern language models and their architectures.

Topics:

- Statistical vs neural language models.
- RNN-based language models.
- Sequence-to-sequence models.
- Attention & Transformer architecture.
- Pre-training vs fine-tuning paradigm.
- Large Pre-trained Models (BERT, GPT, RoBERTa).
- Scaling laws of LLMs.
- Challenges: hallucinations, bias, resource intensity.

Hands-On Activity:

- Train a character-level RNN for text generation.
- Implement a Transformer from scratch (mini version).
- Fine-tune GPT-2 for text generation.

4 Natural Language Processing (NLP)

Session 8: Machine Translation

Goal: Apply deep learning architectures for sequence-to-sequence translation.

Topics:

- Rule-based & statistical MT vs neural MT.
- Encoder-Decoder architecture.
- Attention mechanism for translation.
- BLEU score evaluation.
- Seq2Seq with LSTMs for translation.
- Transformer-based MT (Google's Transformer, MarianMT).
- Multilingual translation challenges.
- Low-resource language translation techniques.

Hands-On Activity:

- Train a Seq2Seq LSTM model for English → French translation.
- Fine-tune MarianMT on custom bilingual corpus.
- Evaluate translations with BLEU score.

Session 9: Speech & Multimodal NLP

Goal: Extend NLP techniques to speech and multimodal understanding.

Topics:

- Speech-to-text pipeline basics (ASR).
- Feature extraction from audio (MFCC, spectrograms).
- Transformer-based ASR (Wav2Vec2).
- Multimodal embeddings (CLIP, Flamingo).
- Audio + text joint models.
- Vision + language tasks (image captioning, VQA).
- Challenges in multimodal alignment.
- Applications: virtual assistants, medical AI, video analytics.

Hands-On Activity:

- Fine-tune Wav2Vec2 for speech recognition.
- Use CLIP for text-to-image similarity.
- Build an image captioning model with CNN + RNN/Transformer.

5

Generative AI Essentials & Prompt Engineering

Session 1: Introduction to Generative AI

Goal: Build a conceptual foundation of Generative AI, its history, evolution, and modern applications.

Topics:

- Generative vs Discriminative models.
- Historical evolution: RBMs → VAEs → GANs → Transformers.
- Applications: text, image, audio, video, multimodal.
- Generative vs Retrieval-based approaches.
- Data requirements & computational challenges.
- Key evaluation metrics (Inception Score, FID, BLEU, ROUGE, perplexity).
- Limitations of generative models (mode collapse, hallucinations).
- Ethical challenges: bias, misinformation, copyright.

Hands-On Activity:

- Generate synthetic images using a pre-trained GAN model.
- Explore text generation using GPT-2.
- Compare discriminative vs generative model outputs.

Session 2: Autoencoders & GANs

Goal: Understand core generative model architectures for unsupervised learning and synthetic data creation.

Topics:

- Autoencoder architecture: encoder, decoder, bottleneck.
- Variational Autoencoders (VAEs) & latent space sampling.
- Applications of autoencoders: denoising, anomaly detection.
- GAN architecture: generator vs discriminator.
- Minimax game in GANs & training instability.
- Advanced GANs: DCGAN, CycleGAN, StyleGAN.
- Common problems: mode collapse, vanishing gradients.
- Evaluation metrics for generative images.

Hands-On Activity:

- Implement an autoencoder for image reconstruction (MNIST).
- Train a simple GAN to generate digits.
- Use StyleGAN to generate realistic human faces.

Session 3: Transformers & Attention Mechanism

Goal: Learn the architecture that powers modern LLMs and multimodal models.

Topics:

- Limitations of RNNs & CNNs for sequence tasks.
- Self-attention mechanism: queries, keys, values.
- Scaled dot-product attention.
- Multi-Head Attention & positional encoding.

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Generative AI Essentials & Prompt Engineering

- Transformer encoder vs decoder.
- Pre-training (masked language modeling, autoregressive).
- Evolution: BERT, GPT, T5.
- Scaling laws for transformers.

Hands-On Activity:

- Implement scaled dot-product attention from scratch.
- Train a small transformer for text classification.
- Visualize attention maps for interpretability.

Session 4: Small Language Models (SLMs)

Goal: Explore lightweight LMs optimized for efficiency, edge computing, and domain adaptation.

Topics:

- Definition of Small Language Models (SLMs).
- Distillation techniques (Knowledge Distillation, TinyBERT, DistilGPT).
- Quantization & pruning for model compression.
- Trade-offs: accuracy vs efficiency.
- SLMs for edge devices & enterprise applications.
- Fine-tuning small LMs for domain tasks.
- Comparison: SLMs vs LLMs.
- Open-source SLMs (Alpaca, Phi-2, Mistral).

Hands-On Activity:

- Fine-tune DistilBERT on a custom text dataset.
- Apply quantization on a transformer model.
- Deploy an SLM on a lightweight container (CPU-only).

Session 5: Prompt Engineering Essentials

Goal: Learn structured prompting techniques for effective interaction with LLMs.

Topics:

- Basics of prompt construction.
- Zero-shot prompting.
- Few-shot prompting.
- Chain-of-thought prompting.
- Instruction tuning & role prompting.
- Handling ambiguity in prompts.
- Prompt evaluation & iteration.
- Failures in prompting & mitigation strategies.

Hands-On Activity:

- Compare zero-shot vs few-shot results on summarization.
- Use chain-of-thought prompting for reasoning tasks.
- Build a structured prompt library for enterprise tasks.

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Generative AI Essentials & Prompt Engineering

Session 6: Advanced Prompting Strategies

Goal: Master advanced prompting frameworks and optimization strategies for production-grade AI apps.

Topics:

- Self-consistency prompting.
- Prompt chaining & multi-step workflows.
- Retrieval-Augmented Prompting with external knowledge.
- Program-aided prompting (PAP).
- Auto-prompting & reinforcement learning for prompts.
- Prompt optimization with LLM feedback loops.
- Guardrails & prompt injection attacks.
- Advanced frameworks: LangChain, Guidance, DSPy.

Hands-On Activity:

- Implement RAG with a vector DB for knowledge-grounded prompting.
- Experiment with self-consistency prompts on reasoning tasks.
- Build a small LangChain pipeline with multi-step prompt chaining.

6

LLM-powered Application Development

Session 1: Introduction to Large Language Models (LLMs)

Goal: Build foundational understanding of LLMs, their architecture, and scaling laws.

Topics:

- Evolution of LLMs: GPT, BERT, T5, LLaMA, Falcon.
- Transformer architecture deep dive: self-attention, positional encoding, multi-head attention.
- Scaling laws: model size vs data size vs compute.
- Pretraining objectives: Causal LM, Masked LM, Seq2Seq.
- Capabilities & limitations: emergent abilities, hallucination, bias.

Hands-On Activity:

- Load pre-trained Hugging Face model (GPT-2 or BERT) and generate text.
- Visualize transformer attention heads.

Session 2: Open Source LLMs & Model Variants

Goal: Explore open-source LLM ecosystems, their strengths, and use-cases.

Topics:

- Survey of popular open-source models: LLaMA 2, Falcon, Mistral, BLOOM, OpenLLaMA.
- Trade-offs: model size vs latency vs performance.
- Quantization & pruning for efficient inference.
- Instruction-tuned vs base models.
- GPU/TPU requirements for serving.

Hands-On Activity:

- Deploy LLaMA-2 (7B) on local GPU or Colab with quantization (GGML/4-bit).
- Compare inference performance across variants.

Session 3: Vector Databases

Goal: Learn how embeddings enable semantic search and memory for LLMs.

Topics:

- Text embeddings: Sentence-BERT, OpenAI Embeddings, InstructorXL.
- Vector similarity search: cosine, dot product, Euclidean distance.
- Vector databases overview: FAISS, Pinecone, Weaviate, Milvus, Chroma.
- Indexing strategies: IVF, HNSW, PQ.
- Scaling vector DBs in production.

Hands-On Activity:

- Generate embeddings with Hugging Face or OpenAI.
- Store and query documents using FAISS or Chroma.
- Build a semantic search over custom text corpus

Session 4: Retrieval-Augmented Generation (RAG) Techniques

Goal: Enable LLMs to access external knowledge with retrieval pipelines.

6

LLM-powered Application Development

Topics:

- RAG architecture: retriever + reader pipeline.
- Chunking strategies & embedding storage.
- Query rewriting & reranking.
- Hybrid retrieval (sparse + dense).
- Advanced techniques: memory-augmented RAG, hierarchical
- RAG.

Hands-On Activity:

- Implement a RAG pipeline with LangChain + FAISS/Chroma.
- Build a document Q&A bot using LLaMA + retrieval.

Session 5: LLM Frameworks & Development

Goal: Master modern LLM development frameworks.

Topics:

- LangChain: chains, agents, memory, tools.
- LlamaIndex: document loaders, index types, retrievers.
- Hugging Face Transformers & PEFT (Parameter Efficient Fine-Tuning).
- Guardrails: input/output validation, prompt templating.
- Agent architectures: ReAct, AutoGPT, LangGraph.

Hands-On Activity:

- Build an agent with LangChain to query a vector DB + call APIs.
- Implement LlamaIndex indexing & query system.

Session 6: LLM Application Development

Goal: Design and build scalable LLM-powered applications.

Topics:

- Prompt engineering patterns: zero-shot, few-shot, chain-of-thought, tree-of-thought.
- Tool usage and multi-agent systems.
- Conversation memory: short-term vs long-term context.
- Context window optimization (sliding windows, summarization).
- API orchestration & async pipelines.

Hands-On Activity:

- Build a customer-support chatbot with LangChain.
- Add conversation memory + multi-turn reasoning.

Session 7: Fine-tuning & Model Adaptation

Goal: Adapt pre-trained models for domain-specific tasks.

6

LLM-powered Application Development

Topics:

- Full fine-tuning vs LoRA vs QLoRA vs adapters.
- Instruction tuning & alignment with RLHF.
- Dataset curation for fine-tuning: cleaning, augmentation, synthetic data generation.
- Cost/performance trade-offs in fine-tuning.

Hands-On Activity:

- Fine-tune a BERT/LLama-2 model on a custom dataset with LoRA.
- Evaluate domain adaptation improvements.

Session 8: Generative AI Application Deployment

Goal: Learn how to deploy LLMs and apps in production.

Topics:

- Deployment strategies: REST API, gRPC, serverless, edge deployment.
- GPU scheduling on Kubernetes (NVIDIA GPU Operator, Triton Inference Server).
- Load balancing for LLM inference (vLLM, DeepSpeed, TensorRT).
- Scaling to millions of users: caching, sharding, async requests.
- MLOps for LLMs (LLMOps): continuous evaluation & retraining.

Hands-On Activity:

- Containerize an LLM with Docker + FastAPI.
- Deploy on Kubernetes with autoscaling.

Session 9: LLM Caching & Routing Techniques

Goal: Optimize LLM inference cost, speed, and accuracy.

Topics:

- Prompt caching strategies (semantic vs exact match).
- KV cache & attention reuse for long conversations.
- Multi-LLM routing: selecting the right model based on task/latency/cost.
- Mixture-of-Experts (MoE) models.
- API-level optimizations for throughput.

Hands-On Activity:

- Implement semantic caching with FAISS + OpenAI embeddings.
- Build a router that switches between GPT-3.5 (cheap) and GPT-4 (accurate).

Session 10: Evaluation, Metrics, and Responsible AI for LLMs

Goal: Establish evaluation frameworks and apply responsible AI practices.

Topics:

- LLM evaluation challenges: subjective correctness, hallucination, bias.

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LLM-powered Application Development

- Metrics: BLEU, ROUGE, METEOR, BERTScore, perplexity, win rate.
- Human evaluation & feedback loops.
- Responsible AI: bias mitigation, toxicity detection, fairness audits.
- Governance & compliance: data privacy, GDPR, model auditing.

Hands-On Activity:

- Evaluate outputs with BLEU/ROUGE vs human scoring.
- Apply AI fairness library (e.g., Fairlearn) on model outputs.



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Agentic AI & Autonomous AI Systems Curriculum

Session 1: Agentic AI Essentials

Goal: Establish core understanding of Agentic AI, autonomous reasoning loops, and design principles.

Topics:

- Defining Agentic AI vs traditional AI applications.
- Core agent loop (perception → planning → action → reflection).
- Differences between chatbots, LLM apps, and agents.
- Role of autonomy and self-directed decision-making.
- Integration with external tools, APIs, and environments.
- Memory in agents: ephemeral vs persistent context.
- Typical enterprise applications (assistants, copilots, automation).
- Risks and challenges: hallucination, control, scalability.

Hands-On Activity:

- Build a minimal Python agent that queries a knowledge base and
- performs an action.

Session 2: Architectures & Design Patterns

Goal: Explore scalable design patterns for building and deploying agentic systems.

Topics:

- Core agent architectures: ReAct, Plan-and-Execute, Reflexion.
- Using reasoning traces for better interpretability.
- Decision trees vs feedback loops in agents.
- Error handling patterns: retries, fallbacks, guardrails.
- Human-in-the-loop workflows for oversight.
- Centralized vs decentralized orchestration.
- Modular vs monolithic agent design.
- Enterprise-grade deployment considerations (latency, cost, compliance).

Hands-On Activity:

- Implement a ReAct agent with retry + error handling.

Session 3: Working with LangChain & LCEL

Goal: Learn to design agents with LangChain's LangChain Expression Language (LCEL).

Topics:

- LangChain core components: tools, chains, agents, memory.
- Introduction to LCEL for declarative agent workflows.
- Composable pipelines with nodes and flow control.
- Tool integration: APIs, DB connectors, external services.
- Persistent memory management across sessions.

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Agentic AI & Autonomous AI Systems Curriculum

- Async execution & concurrency in LCEL pipelines.
- Debugging & tracing agent workflows.
- Best practices for scalable LCEL systems.

Hands-On Activity:

- Build a LangChain LCEL agent that queries APIs and stores results in memory.

Session 4: Building AI Agents with LangGraph

Goal: Use LangGraph for graph-based autonomous agents.

Topics:

- Why graph-based orchestration for agents.
- LangGraph architecture: nodes, edges, transitions.
- Handling cycles in agent workflows.
- Branching logic based on dynamic conditions.
- Adding external API nodes for enhanced actions.
- Fault tolerance & recovery mechanisms in graphs.
- Comparing LangGraph vs LCEL agent workflows.
- Use cases: task planners, conversational flows, workflow automation.

Hands-On Activity:

- Build a LangGraph workflow with decision-based branching.

Session 5: Implementing Agentic RAG

Goal: Combine RAG with agentic workflows for dynamic knowledge retrieval.

Topics:

- Standard RAG vs Agentic RAG architectures.
- Multi-step retrieval with adaptive query reformulation.
- Chunking strategies for dynamic queries.
- Hybrid retrieval (dense + sparse + metadata).
- Evaluating retrieval quality in real-time.
- Adding tool-driven reasoning before/after retrieval.
- Memory-augmented RAG for context persistence.
- Optimizing cost/latency in RAG-enabled agents.

Hands-On Activity:

- Build an Agentic RAG pipeline using LangChain + FAISS/Chroma.

Session 6: Developing AI Agents with Phidata

Goal: Explore Phidata, a Python-first framework for agent development.

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Topics:

- Phidata architecture & design principles.
- Building lightweight Python-first agents.
- Memory persistence across runs.
- API & database integration within agents.
- Orchestrating multi-step workflows with Phidata.
- Multi-modal capabilities: text + structured data.
- Deploying Phidata agents in production environments.
- Comparison with LangChain/LangGraph-based approaches.

Hands-On Activity:

- Build a Phidata agent integrated with a SQL database.

Session 7: Multi-Agent Systems with LangGraph & CrewAI

Goal: Build collaborative & specialized agent teams.

Topics:

- Multi-agent collaboration principles.
- Leader-worker vs peer-to-peer orchestration.
- CrewAI features: team roles, message passing.
- Role specialization in agents (planner, executor, verifier).
- Coordination protocols for multi-agent tasks.
- Conflict resolution & consensus-building.
- Use cases: enterprise workflows, research teams, automation bots.
- Scaling multi-agent systems in production.

Hands-On Activity:

- Implement a multi-agent system with LangGraph + CrewAI roles.

Session 8: Advanced Agent Development with Autogen

Goal: Harness Microsoft Autogen for advanced multi-agent collaboration.

Topics:

- Autogen framework overview.
- Agents as modular programs with specialized roles.
- Defining communication rules between agents.
- Human-in-the-loop integration in Autogen.
- Tool-enabled agents for enterprise tasks.
- Multi-agent conversation orchestration.
- Scaling workflows with Autogen at enterprise scale.
- Observability & debugging in Autogen systems.

Hands-On Activity:

- Build a two-agent collaboration workflow with Autogen.

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Agentic AI & Autonomous AI Systems Curriculum

Session 9: AI Agent Observability & AgentOps

Goal: Enable monitoring, debugging, and governance for agents in production.

Topics:

- Why observability matters in non-deterministic agents.
- Key metrics: tool success rate, latency, reasoning depth.
- Logging & traceability of agent actions.
- Visualization of reasoning chains & workflows.
- Integrating LangSmith for observability.
- AgentOps concepts for lifecycle governance.
- Preventing runaway agents (cost caps, safety nets).
- Monitoring compliance & responsible usage.

Hands-On Activity:

- Monitor and trace an agent using LangSmith.

Session 10: RAG Architecture with Vector DBs

Goal: Deep dive into RAG infrastructure with scalable vector databases.

Topics:

- Vector DB fundamentals: embeddings, indexing, retrieval.
- Popular solutions: FAISS, Pinecone, Milvus, Weaviate.
- Indexing algorithms: HNSW, IVF, PQ.
- Schema design for hybrid search.
- Real-time ingestion for streaming data pipelines.
- Scaling retrieval latency vs accuracy trade-offs.
- Fault tolerance & replication in vector DBs.
- Integration with agent workflows at scale

Hands-On Activity:

- Build a RAG system using Milvus/Pinecone integrated with
- LangChain agents.

Session 11: Advanced RAG Patterns & Evaluation

Goal: Explore cutting-edge retrieval & evaluation strategies.

Topics:

- Hierarchical RAG for multi-level summarization.
- Agent-guided query rewriting.
- Context compression for limited token windows.
- Evaluating RAG pipelines with automated benchmarks.
- Measuring factual consistency & groundedness.
- Balancing retrieval cost vs performance.

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Agentic AI & Autonomous AI Systems Curriculum

- Enterprise evaluation frameworks (GAIA, HELM, RAGAS).
- Deploying feedback loops for continuous RAG improvement.

Hands-On Activity:

- Benchmark multiple RAG patterns and measure evaluation metrics.

Session 12: Continuous Integration for Agentic Systems

Goal: Automate testing, deployment, and lifecycle management of AI agents.

Topics:

- CI/CD pipelines for AI agents.
- Unit testing loops & tool integrations.
- Continuous data validation with DVC/LakeFS.
- Canary deployments for agents in production.
- Versioning models & agent configurations.
- Automating monitoring and rollback strategies.
- Continuous evaluation for drift detection.
- Best practices for secure agent CI/CD pipelines.

Hands-On Activity:

- Set up GitHub Actions to deploy a LangChain agent with automated testing.

Capstone 2: Enterprise-Grade Agentic AI System (Advanced)

Goal: Apply all learned concepts to a real-world project.

Project Requirements:

- Select an enterprise workflow (customer support, finance copilot, knowledge worker).
- Build multi-agent collaboration (LangGraph + CrewAI/Autogen).
- Integrate RAG with vector DB.
- Add observability (LangSmith/AgentOps).
- Deploy with CI/CD on cloud (AWS/GCP/Azure).
- Implement governance: safety nets, monitoring dashboards.
- Optimize cost-performance with caching & routing.
- Deliver end-to-end working solution with documentation.

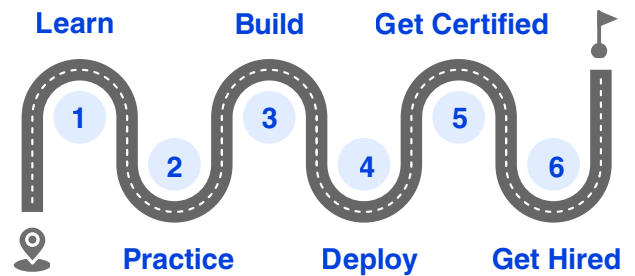


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